

Vatsal Sinha

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EDUCATION

Rheinland-Pfälzische Technische Universität Kaiserslautern-Landau

MS in Computer Science - Specialization in AI/ML

Kaiserslautern, DE

Apr 2026 – Present

PES University

B.Tech in Computer Science

Bangalore, IN

Aug 2018 – May 2022

EXPERIENCE

Software Engineer II

Akamai Technologies

Mar 2025 – April 2026

Bangalore, IN

- Architected a React/FastAPI DID system automating **12k+ phone numbers** across **30+ countries**, slashing overhead by **97%** while implementing **LDAP-based RBAC** for secure allocation and management.
- Optimized a global configuration platform with enhanced validation and templates, reducing configuration errors by **80%** and significantly boosting search UI performance by **30%** for cross-functional engineering teams.
- Engineered a self-service tool for automated role elevation, eliminating manual support bottlenecks and decreasing global ticket volume by **92%** while maintaining strict enterprise-level security and compliance standards.
- Automated enterprise-wide password rotation using **Ansible** for **20+ servers**, ensuring **100% security compliance** and reducing rotation-related system downtime by **95%** across all production and dev environments.

Software Engineer I

Akamai Technologies

Jan 2022 – Mar 2025

Bangalore, IN

- Engineered an enterprise release management platform (React, Node.js) integrating **Jira and Salesforce** to streamline workflows, reducing release cycle friction by **35%** and improving auditability for **100+** global teams.
- Architected a network automation platform (React, Flask) to orchestrate datacenter device upgrades via **NetMRI**, automating pre-check scripts for **1,000+ devices** and slashing manual upgrade effort by **80%** across the fleet.
- Orchestrated **5+ multibranch CI/CD pipelines** using Jenkinsfile and Docker to automate complex enterprise deployments, achieving **100% build consistency** and reducing production deployment time by **65%**.
- Integrated **BlackDuck** and **SonarQube** into automated security workflows to detect vulnerabilities early, reducing critical production vulnerabilities by **60%** and ensuring **100%** adherence to internal security compliance standards.

PROJECTS

Research Publication: AI-Assisted College Recommendation System | *AI/ML, Python*

August 2022

- Published peer-reviewed research in **Springer Nature** (LNNS Vol. 458) detailing a hybrid recommendation engine that optimized college selection accuracy by **15%** for aspiring undergraduate students.
- Engineered an automated data acquisition pipeline via web scraping to aggregate profiles for **1,000+ institutions**, mapping datasets against user profiles built through explicit feature extraction and NLP analysis.
- Implemented a novel **hybrid recommendation architecture** to match student interests with academic data, delivering personalized results with an **89% user satisfaction rate** across diverse global academic datasets.

Image Classification App | *Django, React, TensorFlow, Keras*

August 2024 – October 2024

- Engineered a **TensorFlow CNN** with **94.2% accuracy** and **0.12 loss** for model optimization.
- Developed a **Django API** with **sub-200ms latency** using async pipelines for real-time image preprocessing.
- Built a **React UI** for **500+ uploads** displaying **Top-3 confidence scores** with real-time inference feedback.

TECHNICAL SKILLS

Languages: C/C++, Python, SQL, JavaScript, HTML/CSS, R, Bash

Frameworks/Tools: React, Node.js, Django, Flask, Express, PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy, SciPy, Git, Docker, AWS, Jupyter Notebook, Excel, PostgreSQL, MongoDB, Firebase, Redis, Computer Vision, NLP, CI/CD, Agile, DevOps, NoSQL, Linux